KIOXIA

KIOXIA XG6 Series (M.2)

Client NVMe™ SSD

KIOXIA XG6 Series utilizes KIOXIA latest 96-layer, 3D TLC (3-bit-per-cell) flash memory. With 4th generation BiCS FLASH™ and SLC cache features, the XG6 Series SSDs reach up to sequential read/write speeds of 3180 MB/s and 2960 MB/s respectively and deliver up to 355,000 random read and 365,000 random write IOPS. In addition to high performance, the XG6 Series carries on the low power design of the XG family, consuming 4.7 W or less in active mode and less than 3 mW in stand-by mode.

The new XG6 Series is optimized for power-sensitive mobile PCs, performance-oriented gaming PCs, as well as data center environments for server-boot, caching and logging.
Available in a compact M.2 2280 single-sided form factor, the XG6 Series comes in three capacity models of 256 GB, 512 GB and 1024 GB, each with the option of a Self-Encrypting Drive (SED) model supporting TCG Opal Version 2.01.

Product image may represent a design model.

Key Features

- KIOXIA 96-Layer BiCS FLASH™
- PCIe® Gen3 x4, NVMe™
- Capacities up to 1,024 GB
- M.2 2280 Single-sided
- TCG Opal 2.01 Optional for SED

Key Applications

- Thin performance notebook PCs
- High-performance desktop PCs
- Gaming PCs
- Server-boot, caching & logging use in data center

Specifications

Base Model Number	**KXG60ZNV1T02**	**KXG60ZNV512G**	**KXG60ZNV256G**
SED Model Number	**KXG6AZNV1T02**	**KXG6AZNV512G**	**KXG6AZNV256G**
Capacity	1,024 GB	512 GB	256 GB
Basic Specifications			
Form Factor	M.2 2280-S2 Single-sided		
Interface	PCIe® 3.0, NVMe™ 1.3a		
Maximum Interface Speed	32 GT/s (PCIe® Gen3 x4)		
Flash Memory Type	BiCS FLASH™ TLC		

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Specifications (Continued)

Capacity	1,024 GB	512 GB	256 GB
Performance (Up to)			
Sequential Read	3,180 MB/s	3,100 MB/s	3,050 MB/s
Sequential Write	2,960 MB/s	2,800 MB/s	1,550 MB/s
Random Read	355K IOPS	325K IOPS	270K IOPS
Random Write	365K IOPS	355K IOPS	335K IOPS
Power Requirements			
Supply Voltage	3.3 V ±5 %		
Power Consumption (Active)	4.7 W typ.	4.1 W typ.	4.0 W typ.
Power Consumption (L1.2 mode)	3.0 mW typ.		
Reliability			
MTTF	1,500,000 hours		
TBW	600	300	150
Dimensions			
Thickness	2.23 mm Max		
Width	22.0 mm ± 0.15 mm		
Length	80.0 mm ± 0.15 mm		
Weight	7.3 g Max		7.0 g Max
Environmental			
Temperature (Operating)	0 °C to 95 °C (Controller Temperature)		
Temperature (Operating)	0 °C to 85 °C (Other Components Temperature)		
Temperature (Non-operating)	-40 °C to 85 °C		
Humidity (Operating)	0 % to 90 % R.H.		
Vibration (Operating)	196 m/s^2 { 20 Grms } (20 Hz to 2,000 Hz)		
Shock (Operating)	14.7 km/s^2 { 1,500 G } (0.5 ms)		

Availability of the SED model line-up may vary by region.

Definition of capacity: KIOXIA Corporation defines a megabyte (MB) as 1,000,000 bytes, a gigabyte (GB) as 1,000,000,000 bytes and a terabyte (TB) as 1,000,000,000,000 bytes. A computer operating system, however, reports storage capacity using powers of 2 for the definition of 1 GB = 2^30 = 1,073,741,824 bytes and therefore shows less storage capacity. Available storage capacity (including examples of various media files) will vary based on file size, formatting, settings, software and operating system, and/or pre-installed software applications, or media content. Actual formatted capacity may vary.

MTTF (Mean Time to Failure) is not a guarantee or estimate of product life; it is a statistical value related to mean failure rates for a large number of products which may not accurately reflect actual operation. Actual operating life of the product may be different from the MTTF.

TBW: Terabytes Written. The number of terabytes that may be written to the SSD for the specified lifetime.

Read and write speed, tested on the state of "SLC cache=ON", may vary depending on the host device, read and write conditions, and file size.

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